



RAFFLES INSTITUTION

H2 Computing (9569)

2026 Year 6

Chapter S4F: Networks

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1. What is the Internet

A **computer network** consists of two or more computing devices connected via a transmission medium to enable communication, information exchange, and resource sharing. Networks can be categorised based on their geographical scale, the medium used for transmission, and their organisational structure.

The Internet is a global network of networks that enables computers worldwide to interconnect and communicate. It originated in the 1960s from a US military project, which created ARPANET (Advanced Research Projects Agency Network). This prototype allowed research teams in different locations to share data and collaborate efficiently without needing to travel physically.

1.1 Physical Structure

The physical infrastructure of the Internet relies on high-capacity backbone cables, primarily made of fiber optics, which span across continents. These cables are laid along sea beds as trans-continental links, forming the core network. Fiber optics allow data to travel vast distances at high speeds with very little loss of signal quality.

Internet Service Providers (ISPs) connect directly to this global backbone. Large national ISPs then distribute the connection to smaller regional providers, who ultimately deliver internet access to homes and businesses.

In areas where physical cables are difficult or expensive to install, satellite communication provides an alternative. Data signals are relayed via orbiting satellites, making it possible to access the internet in remote or rural locations where terrestrial infrastructure is not available.



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1.2 Uniform Resource Locators (URLs)

A URL is the unique address used to locate a resource on the Internet. It typically follows this structure:

`http://www.domainname.com/folder/subfolder/webpage.html#section`

Component	Example	Function
Protocol	<code>http://</code>	Specifies the rules/method used to transfer the resource.
Hostname	<code>www</code>	Identifies the specific server or computer on the network.
Domain Name	<code>domainname.com</code>	The registered address of the website.
Path	<code>folder/subfolder/</code>	Shows the directory location of the file on the server.
Filename	<code>webpage.html</code>	The specific name and type of document accessed.
Fragment	<code>#section</code>	Optional. Refers to a specific section or position <i>within</i> the webpage.



1.3 Internet Registries and Registrars

Internet Registries are organisations that manage critical internet resources, ensuring uniqueness and orderly operation. There are two main types:

1. IP Address Registries

- **IANA (Internet Assigned Numbers Authority):** Manages global allocation of IP addresses.
- **RIRs (Regional Internet Registries):** Distribute IP blocks to ISPs within specific geographical regions (e.g., APNIC for Asia Pacific).
- **ISPs (Internet Service Providers):** Assign specific IP addresses to end-users/devices.

2. Domain Name Registries

- Manages the master database for a specific Top-Level Domain (TLD), e.g., .com, .sg.
- Ensures all domain names within that TLD are unique.
- Example: **SGNIC** (Singapore Network Information Centre) is the registry for .sg domain names.

Registrars

- Companies accredited to sell domain names directly to the public.
- Act as the middleman: They collect customer details and submit them to the Registry.
- Maintain records of ownership and hosting details.

The Registry owns the central database and sets the rules, whereas the Registrar is the retail agent that interacts directly with customers.



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1.4 Internet Protocol Address (IP Address)

An IP address is a unique identifier assigned to a device on a network. It functions like a postal address, specifying the source and destination location for data packets.

IPv4 vs IPv6

Feature	IPv4	IPv6
Size	32 bits	128 bits
Format	4 decimal numbers (0-255) separated by dots. Each number can range from 0 to 255 (inclusive) and correspond to one byte (8 bits). e.g., 192.168.1.1	8 groups of 4 hexadecimal digits separated by colons. For compactness, leading zeros and up to one consecutive sequence of zero-only groups may be omitted. e.g., 2001:DB8::20:17:bad:c0de
Total Addresses	Approx. 4.3 Billion (4,294,967,296)	3.4×10^{38} (Effectively unlimited)
Reason for Change	Address exhaustion due to the massive growth of internet-enabled devices.	Designed to solve IPv4 address exhaustion and improve network efficiency and security support.
Example		

Special IPv4 Addresses

- **127.0.0.1**: Known as **localhost**. Refers specifically to your own computer. Used for testing.
- **0.0.0.0**: Represents an unspecified or unknown address.



Structure of an IP Address

An IPv4 address is hierarchical and split into two logical parts:

1. **NetID:** Identifies the network. Used for routing data between different networks.
2. **HostID:** Identifies the specific device *within* that network. Only checked once the data arrives at the destination network.

Classful Addressing

Originally, addresses were divided into fixed classes. The first few bits determine the class:

Class	Starting Bits	NetID Bits	HostID Bits	Meaning
A	0	7	24	Few networks, many hosts (up to 16 million)
B	10	14	16	Medium networks (up to 65,000 hosts)
C	110	21	8	Many networks, few hosts (only 256 hosts)
D	1110	-	-	Multicasting
E	1111	-	-	Reserved

The system was too rigid. You could not split a network ID: one network ID = one physical LAN.

Consider a company with **150 computers** spread across **6 departments** (6 separate LANs).

- In the past, they had to apply for **6 different class C IDs**.
- Each Class C provides 256 addresses.
- **Total assigned:** $6 \times 256 = 1536$ addresses.
- **Actual usage:** Only 150 addresses.
- **Result:** 1386 addresses were wasted and unavailable to others.

Classful addressing is no longer used in modern networks but is included for historical understanding.



Modern Solutions

1. CIDR (Classless Inter-Domain Routing)

- Removes the strict boundaries of Classes A, B, C.
- Allows the division between NetID and HostID to be flexible.
- ISPs can now allocate exactly the number of addresses an organisation actually needs, preventing waste.

2. Subnetting

Subnetting is the process of dividing a single network into smaller logical networks by borrowing bits from the host portion of the IP address to create a subnet identifier.

Example:

- A Class C network (e.g., 193.10.9.0/24) has
 - 24 bits for the network portion
 - 8 bits for the host portion
- If 3 bits are borrowed from the host portion:
 - Number of subnets = $2^3 = 8$ subnets
 - Remaining host bits = 5 bits
 - Number of usable hosts per subnet = $2^5 - 2 = 30$ hosts

Example of 8-bit host portion:	0	0	1	0	1	0	0	1
--------------------------------	---	---	---	---	---	---	---	---

First 3 bits = Subnet ID, Last 5 bits = Host ID

In each subnet, two addresses cannot be assigned to devices:

- **Network Address:** Identifies the subnet itself (all host bits = 0)
- **Broadcast Address:** Used to send data to all devices in the subnet (all host bits = 1)

1.5 Domain names and Domain Name System (DNS)

A domain name identifies the location or site of a resource on the Internet. Each location has a unique domain name. Once registered, domain names are stored in publicly accessible databases.

Domain names are structured in a hierarchy, written as a string separated by full stops.

- **TLDs (Top Level Domains):**
 - **Generic TLDs:** .com, .edu, .org
 - **Country TLDs:** .sg, .uk, .my
- **2LDs (Second Level Domains):** e.g., google, yahoo
- **3LDs (Third Level Domains):** e.g., www, mail, docs



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A complete address specifying the exact location (e.g., www.google.com) uniquely identifies resources and allows DNS to resolve them to IP addresses.

Each domain name can correspond to one or more IP addresses. This allows for load balancing or redundancy, where multiple IP addresses can be associated with a single domain name.

How DNS Works

The DNS operates using a **distributed hierarchical structure**. It manages the mapping between domain names and IP addresses. Each domain is associated with authoritative DNS servers that store the domain name to IP address mappings.

The Process:

1. Browser sends a request to the local DNS server.
2. Local DNS server checks if it has the IP address:
 - If found in cache → return immediately.
3. If not found, it queries higher-level servers:
 - Root DNS server → directs to the appropriate TLD server (.com, .sg, etc.)
 - TLD server → directs to the authoritative DNS server for the domain
4. Authoritative server returns the IP address.
5. Local DNS caches the result and returns it to the browser.
6. Browser uses the IP address to connect to the web server.

DNS allows domain names to be used instead of IP addresses, making websites easier to access.



1.6 Static and Dynamic IP Addresses

Static IP Addresses

Some computers are connected to the Internet at all times and always use the same address. These are typically **servers, routers, or infrastructure devices** that require a consistent and fixed network identity.

Dynamic IP Addresses

Most end-user devices connecting through an ISP are assigned **dynamic IP addresses**. This is managed by the ISP's **DHCP (Dynamic Host Configuration Protocol) server**. When a device connects to the network, the DHCP server automatically assigns an available IP address from a pool. This address might be different each time you connect. This allows ISPs to manage their resources efficiently. They can serve many customers with a limited set of addresses, since not everyone is online at the same time.

IP Type	Advantages	Disadvantages
Static	• Reliable: Fixed address simplifies remote access, hosting, and configuration. • Secure: Easier to implement strict access control (e.g., configure firewalls to allow traffic only from specific trusted IPs).	• Cost: ISPs often charge extra for a static IP. • Management: Manual configuration is time-consuming and harder to maintain.
Dynamic	• Cost-effective: Efficient use of IP addresses reduces costs for ISP and user. • Easy & Scalable: Automatic setup; new users can connect without manual configuration.	• Inconsistent: Changing IPs can disrupt services that require a fixed address. • Troubleshooting: Diagnosing network issues is slightly harder when the address keeps changing.



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2. LAN and WAN

Based on the geographical coverage of the network, a network can be classified into Local Area Networks (LANs) and Wide Area Networks (WANs).

LAN

In a LAN, computer systems are situated relatively close to each other, for example in the same building or cluster of buildings such as a school.

- **Connection:** Uses wire cables, wireless links (e.g., Wi-Fi) or fibre optic cables.
- **Sharing:** Allows sharing of data files and hardware resources such as printers and scanners.
- **Structure:** Usually follows a **Client-Server model**, where **clients** are user computers and **servers** provide central storage and services.
- **Features:** Fast data transfer, high reliability, lower setup cost, easier maintenance and security. However, scalability can be an issue for very large networks.
- **Ownership:** Usually owned and operated by a single organisation.

WAN

A WAN connects computers in geographically dispersed locations, enabling communication across large areas such as cities, countries or continents.

- **Connection:** Uses public infrastructure like telephone links, undersea cables and satellites.
- **Features:** Higher costs, generally slower speeds, and more complex security management due to public connections.
- **Purpose:** Often used to link multiple separate LANs together, such as in a multinational organisation.



3. Intranet

An Intranet is a private network that operates exclusively within an organisation. It uses the same technology as the Internet but is restricted to authorised users only.

Key Features

- **Access Control:** Limited to employees or members, usually protected by passwords or user accounts.
 - Different levels of access rights/passwords can be used to ensure that only specific people can access specific facilities or data.
- **Services:** Provides internal web pages, email, file sharing, and collaboration tools.
- **Content Quality:**
 - As there is a smaller volume of content compared to the Internet, it is more relevant to the organisation.
 - Information is more likely to be correct, relevant and up-to-date.
- **Security:** Isolated from the public internet, reducing risks of external attacks. Often protected by firewalls.
- **Performance:** Generally faster and more reliable due to higher bandwidth and less traffic congestion.

Connection Structure

- Built using the organisation's internal LANs (Local Area Networks) and WANs (Wide Area Networks).
- Remote access is securely enabled via a VPN (Virtual Private Network).
- When parts of the Intranet are made available to selected outside users (e.g., partners, suppliers), this is called an **Extranet**.

Virtual Private Network (VPN)

A technology that creates a secure, encrypted "tunnel" over a public network (like the Internet).

Purpose

- Allows users to access the Intranet safely from outside the office (e.g., working from home or travelling).
- Protects data transmission by encrypting it, preventing unauthorised people from reading or intercepting the information.



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Example: National Museum

The museum has multiple branches in different cities.

Network Setup:

- Within each branch, computers and devices are connected via a **LAN**, enabling fast local file sharing and resource access.
- All branches are linked together through a **WAN**, forming a unified private network across the entire organisation.

Role of Intranet:

- Acts as a central platform for staff to access internal information, such as artifact databases, exhibition plans, and administrative documents.
- Ensures **security**: Only authorised employees can log in, protecting sensitive data from public access.
- Improves **efficiency**: Allows seamless collaboration between departments and locations, with faster data transfer speeds than the public internet.

Remote Access: Staff working off-site can connect securely to the Intranet using a **VPN**, ensuring they can access necessary resources anytime, anywhere.

4. Communication and Transmission

4.1 Data flow modes

Data flow along a link can be simplex, half duplex or full duplex.

- **Simplex:** Data flows in **one direction only**.
 - Example: TV broadcasting, Keyboard to computer.
- **Half Duplex:** Data flows in **both directions but not at the same time**. Devices take turns sending and receiving.
 - Example: Walkie-talkie.
- **Full Duplex:** Data flows in **both directions simultaneously**. Requires separate channels for sending and receiving.
 - Example: Mobile phone calls, Modern Ethernet.



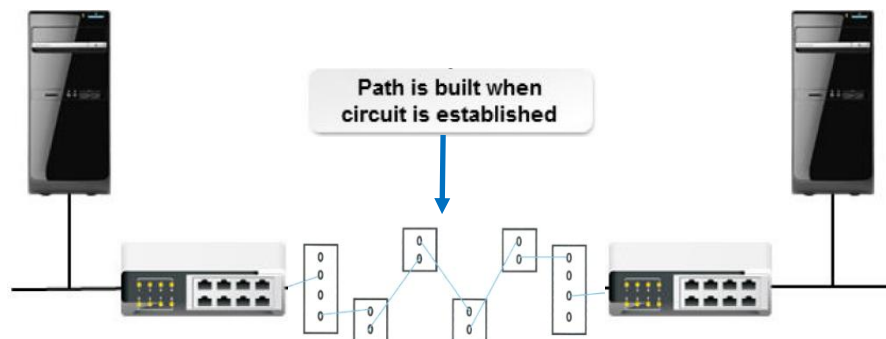
4.2 Message types

- **Broadcast:** One-to-all communication. Message sent to **every device** on the network.
 - Simple but inefficient; wastes bandwidth.
- **Multicast:** One-to-many communication. Message sent only to devices that have **subscribed/joined a specific group**.
 - Efficient way to send data to selected recipients.
- **Unicast:** One-to-one communication. Message sent directly to **one specific device**.
 - Used for private and direct communication.

4.3 Circuit Switching

A dedicated communication path is set up between two devices before data transfer begins.

- Bandwidth is reserved exclusively for the duration of the connection, even if no data is being sent, which can lead to inefficient use of resources.
- Data is transmitted in order and follows the same path throughout the connection.
- Example: Traditional telephone systems (Public Switched Telephone Network).





4.4 Packet switching

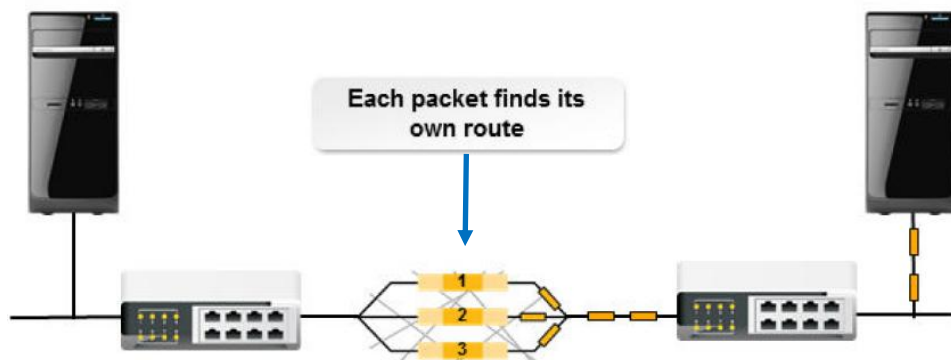
Data is broken into small chunks called **packets**. Each packet travels independently and can take **different routes** depending on traffic.

Connectionless Service

- No setup required. Packets are sent without establishing a connection.
- Packets can arrive out of order or get lost. No guarantee of delivery.
- Fast and low overhead.
- **Protocol:** UDP
- **Used for:** Live streaming, online gaming, VoIP (speed more important than perfection).

Connection-oriented Service

- Virtual circuit established before transmission.
- Reliable delivery using acknowledgements, sequencing, and retransmission.
- Packets may arrive out of order, but are reassembled into the correct order at the receiver.
- Slower due to error checking and connection setup.
- **Protocol:** TCP
- **Used for:** Web browsing, file transfer, email (accuracy is prioritised over speed).





4.5 Data Packet

Data to be transmitted across a network is broken down into manageable chunks called **packets**. The size of each packet can be fixed or variable. Each packet contains:

- **Header:** Control information.
- **Payload:** The actual data being sent.
- **Trailer:** Error checking information.

Error Detection

Some packets include a trailer containing a checksum or Cyclical Redundancy Check (CRC), used to detect transmission errors and ensure data integrity.

- A value is calculated based on the data at the sender's end and added to the trailer.
- At the receiver's end, the same calculation is performed on the received data.
- If values match: Data is likely correct.
- If values differ: Data is corrupted, and the packet is requested again.

TCP provides reliable transmission using error detection and retransmission mechanisms. UDP does not include these features, making it faster but less reliable.

Parity Check

A simpler method where an extra bit is sent with each byte.

- **Even Parity:** Ensures the total number of 1-bits is even.
- **Odd Parity:** Ensures the total number of 1-bits is odd.

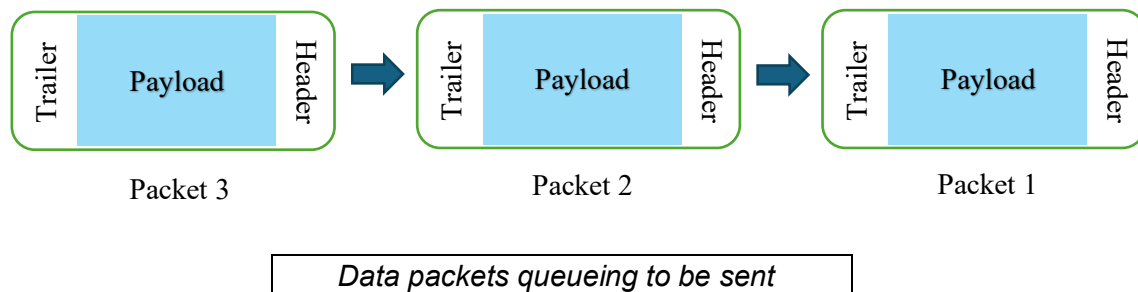
Note: Parity check only detects errors; it cannot fix them.

Packet Header Contents

The header includes:

- Sender's and recipient's **IP addresses**.
- The **protocol** used (e.g., TCP or UDP).
- **Sequence number:** Identifies the order of packets so they can be reassembled correctly at the destination.
- **Time To Live (TTL):** Limits the lifespan of a packet. Each router decreases this value by 1, and the packet is discarded if it reaches zero to prevent infinite loops.

Once all packets are received, they are reassembled in the correct order and the original data is extracted.



4.6 Routers

Routers are used to connect at least two networks, commonly two LANs or WANs, or to connect a LAN and an ISP's network. They serve as intermediate devices that facilitate the transfer of data packets between networks. Each time a packet moves from one router to the next, it is called a hop.

Routers read the recipient's IP address in each packet and forward it via the fastest, least congested route to the next router, repeating the process until the packet reaches its destination. They use routing tables to store device locations and efficient paths, which are updated dynamically to reflect changes in network topology or connectivity. A routing algorithm calculates the optimum path.

When connected to the Internet, the router's external IP address must be globally unique and registered.

4.7 Gateways

Routing packets between networks that use the same protocol (e.g., TCP/IP) is done by a router. However, if the networks use different protocols or data formats, a gateway is required instead.

Gateways function by:

- Stripping away the original header and trailer.
- Translating the data into a format understood by the destination network.
- Repackaging the raw data with new headers before sending it onward.



4.8 Media Access Control (MAC) addresses

Every network device requires a Network Interface Card (NIC) to connect to a network. Each NIC has a unique MAC address, which is assigned and hardcoded by the manufacturer to permanently identify the device.

- **Length:** 48 bits long.
- **Format:** Written as 12 hexadecimal digits, e.g., 38-68-93-B3-07-D0.
- **Purpose:** Used for identifying devices and transferring data within a local network (LAN).

MAC address identifies the specific hardware, whereas IP address identifies its logical location on the network.

5. TCP/IP Model

It is important to have protocols and standards to ensure the proper transfer of data between devices on networks. Protocols specify the format of the data, and the signals required to start, control, and end the transfer. For example, HTTP (HyperText Transfer Protocol) is used by web browsers to request and display web pages.

TCP/IP is a standardised set of protocols used worldwide, enabling communication between devices on different networks regardless of their location. It defines how data should be packetised, addressed, transmitted, routed, and received. This standardisation allows devices to communicate without the need for protocol conversion and supports global connectivity across the Internet.

The TCP/IP protocol stack is organised into four layers:

- Application layer
- Transport layer
- Network layer
- Data Link layer

Each layer has a specific role, and multiple protocols operate within each layer. During communication, data is passed down the stack, and each layer adds its own header (and sometimes a trailer) in a process known as **encapsulation**, forming progressively larger data units. At the receiving end, this process is reversed through **decapsulation**, where each layer removes its own header. This is similar to a game of “pass the parcel”, where each layer adds or removes its own wrapper before the message reaches its destination.



5.1 Application layer

This layer sits at the top of the protocol stack and provides network services to user applications, such as web browsing or email. The application layer includes protocols such as HTTP.

Common protocols:

- HTTP / HTTPS: For web browsing.
- FTP: For file transfer.
- SMTP: For sending email.
- POP3 / IMAP: For receiving email.

Let's see how this sample data is being sent using HTTP through the different layers:
`"Aim for the moon. If you miss, you may hit a star."`

5.2 Transport layer

This layer establishes end-to-end communication. It uses mainly two protocols:

1. TCP (Transmission Control Protocol)

- Splits data into segments/packets.
- Adds **Packet/Sequence Numbers** to ensure correct ordering.
- Adds **Port Numbers** (e.g., Port 80 for HTTP) to ensure the data reaches the correct application.
- Ensures reliability: Requests retransmission if packets are lost and sends acknowledgements.

Packet 1
<code>"Aim for the moon."</code>
Port: 80

Packet 2
<code>"If you miss,"</code>
Port: 80

Packet 3
<code>"you may hit a star."</code>
Port: 80

2. UDP (User Datagram Protocol)

- A simpler, connectionless protocol.
- Has **less overhead** (smaller header) than TCP, making it faster.
- **Unreliable**: No acknowledgement of receipt and no retransmission.
- Used for **Video Streaming, VoIP, Online Games** where speed is more important than perfection.



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5.3 Network layer (Internet Layer)

Also known as the IP layer, this layer determines the path packets take between networks.

- Adds **Source IP Address** and **Destination IP Address**.
- Routers operate at this layer to route packets across different networks.
- IP addresses are used to identify and locate devices/computers across different networks.
- Port numbers are used to identify the specific application or process running on that device.
- Together they form a socket, e.g., 113.10.64.21:80

20.55.5.228 113.10.64.21
Packet 1
"Aim for the moon."
Port: 80

20.55.5.228 113.10.64.21
Packet 2
"If you miss,"
Port: 80

20.55.5.228 113.10.64.21
Packet 3
"you may hit a star."
Port: 80

5.4 Data Link layer

Deals with the physical transmission and hardware (computers/routers) addressing.

- Adds **Source MAC Address** and **Destination MAC Address**.
- MAC addresses are hardcoded hardware addresses.
- The destination MAC address identifies the next hop on the local network. It refers to the physical device receiving it next, not the final destination device.

38-68-93-B3-07-D0 00-0C-29-6C-F3-E5
20.55.5.228 113.10.64.21
Packet 1
"Aim for the moon."
Port: 80

38-68-93-B3-07-D0 00-0C-29-6C-F3-E5
20.55.5.228 113.10.64.21
Packet 2
"If you miss,"
Port: 80

38-68-93-B3-07-D0 00-0C-29-6C-F3-E5
20.55.5.228 113.10.64.21
Packet 3
"you may hit a star."
Port: 80

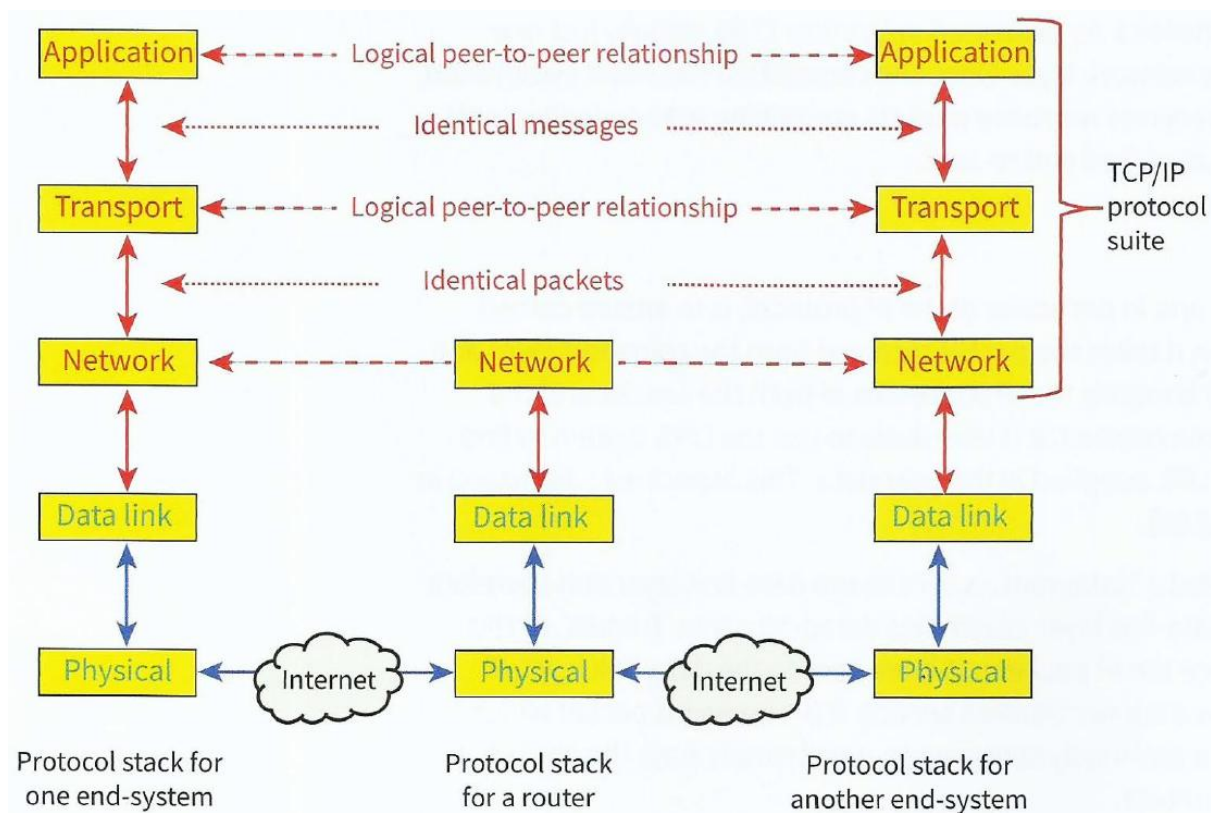


How it works at the receiving end:

As the packet travels up the stack, each layer **removes** its own header and trailer (decapsulation):

1. **Data Link Layer** removes MAC addresses.
2. **Network Layer** removes IP addresses.
3. **Transport Layer** removes Port numbers and reassembles data in order.
4. **Application Layer** presents the final message to the user.

As packets pass through routers, the MAC addresses are stripped and re-added each time. However, the source and destination IP addresses remain the same throughout the entire journey.



While we draw the physical medium separately, in the TCP/IP model, the standards for cables, connectors, and signals (such as electrical voltages, light pulses, or radio waves) are generally considered to be part of the Data Link layer.



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Summary of layers:

Layer	Main Purpose	Address Used
Application	Provides user services	None
Transport	Reliable delivery & port management	Port Number
Network	Routing & logical addressing	IP Address
Data Link	Local delivery & physical addressing	MAC Address

6. Client-Server Architecture

6.1 Client-Server Networking

In a client-server network, one or more **clients** send requests to a central, powerful computer called the **server**. The server processes these requests and provides resources or data to the clients. Clients may store some files locally but can also access shared resources managed by the server. Access is usually controlled via authentication, such as usernames and passwords.

In large networks, different servers handle specific tasks:

Server Type	Function
File Server	Stores and manages files accessible to clients.
Print Server	Manages print jobs and directs them to printers.
Web Server	Delivers web pages and content to clients.
Mail Server	Manages sending and receiving of emails.
Database Server	Stores, organises, and secures data.



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Advantages

- **Security:** Centralised storage allows strict control over access rights.
- **Backup & Recovery:** Data is backed up centrally, making recovery easier.
- **Resource Sharing:** Hardware, software, and data can be shared efficiently.
- **Management:** Clients can be upgraded or replaced easily without affecting the network.

Disadvantages

- **Cost:** Server hardware, software, and maintenance are expensive.
- **Performance:** High traffic can overload the server, causing delays.
- **Single Point of Failure:** If the server fails, the entire network becomes unavailable.

In a **peer-to-peer (P2P) network**, all computers have equal status and can act as both clients and servers. Each device shares its own resources directly with other devices without the need for a central server. This makes setup simple and low-cost, but it is less secure and harder to manage in larger networks compared to a client-server model. The most common use is in file sharing and content distribution.

6.2 Cloud Computing

Cloud computing delivers computing services such as storage, software, and processing power over the internet. It follows the client-server model, allowing users to access resources on demand without needing local hardware or infrastructure.

Service Models

- **Software as a Service (SaaS):** Users access applications via a web browser or interface. The software and data are hosted and managed by the provider.
- **Infrastructure as a Service (IaaS):** Provides virtualised computing resources such as servers and storage.
- **Platform as a Service (PaaS):** Offers a platform for developing, testing, and managing applications.

Advantages

- **Scalability:** Resources can be increased or decreased as needed, and users pay only for what they use.
- **Accessibility:** Data and applications can be accessed from any device and location.
- **Cost Efficiency:** Eliminates the need for expensive hardware and reduces maintenance costs.
- **Collaboration:** Allows multiple users to work on the same data simultaneously.
- **Reliability:** Provides built-in backup and disaster recovery solutions.

Data is stored on remote servers, so it is important to use reputable providers with strong security measures.



6.3 Data Processing

In this model, a client sends a request to a server, which processes it and returns a response. The choice of where to process depends on task complexity, device capability, and security needs. Modern applications often use both methods.

Client-side Processing

Occurs on the user's device (e.g., within a web browser).

- **Lightens load:** Handles simple tasks to reduce server workload.
- **Responsive:** Provides faster interactivity and immediate feedback.
- **Efficient:** Reduces the amount of data transferred over the network.
- **Validation:** Performs initial checks to prevent invalid data from being sent.

Server-side Processing

Occurs on the central server.

- **Powerful:** Efficiently handles complex calculations and large datasets.
- **Data Management:** Directly accesses, queries, and updates databases.
- **Secure:** Enforces strict security and protects sensitive information.

Comparison

Client-side Processing	Server-side Processing
1. Performs initial validation	1. Provides further & secure validation
2. Creates interactivity	2. Queries databases
3. Manipulates user interface	3. Updates server data
4. Reduces load on the server	4. Performs complex calculations
5. Reduces network traffic	5. Generates content / Returns data
6. Prevents basic malicious input	6. Keeps sensitive data secure



6.4 Thin vs Thick-Client Computing

"Thickness" refers to how much processing and storage happens on the client device versus the server.

Thin Client

- **Minimal hardware:** Relies entirely on the server for OS, applications, and data.
- **Centralised management:** Easy to update and secure.
- **Low cost:** Cheaper hardware.
- **Network dependent:** Requires a stable connection to work.
- *Example:* Terminals in call centres, remote desktops.

Thick Client

- **Standalone machine:** Has its own OS, processing power, and storage.
- **High performance:** Can run demanding software.
- **Offline use:** Can function without a network connection.
- **Higher cost:** More expensive hardware and maintenance.
- *Example:* Personal laptops, desktop PCs.

Comparison

Feature	Thin Client	Thick Client
Processing Power	Relies on server	High local capability
Storage	On server	Local storage
Network Need	Essential	Not always required
Cost	Lower hardware cost	Higher hardware cost
Management	Easy, centralised	More complex
Security	Better (data on server)	Data stored locally



7. Networking Summary and Device Comparison

7.1 Intranet vs Internet

Feature	Intranet	Internet
Access	Private, restricted to organisation	Public, global access
Security	High (firewalls, authentication)	Lower, exposed to threats
Purpose	Internal communication and resource sharing	Global information sharing
Users	Employees or authorised members	Anyone

7.2 Network Devices

Device	Function
Modem	Converts digital signals to analogue and analogue signals to digital for internet transmission
Network Interface Card (NIC)	Allows a device to connect to a network and contains a unique MAC address
Router	Routes packets between networks using IP addresses
Gateway	Connects networks using different protocols
Switch	Forwards data to the correct device within a LAN using MAC addresses
Hub	Legacy device that broadcasts data to all devices in a LAN, causing collisions and inefficiency
Bridge	Legacy device that connects LAN segments and filters traffic based on MAC addresses, largely replaced by switches

Hubs and bridges are legacy devices and rarely used in modern switched networks. Modern networks primarily use switches instead of hubs and bridges due to improved efficiency and reduced network collisions.



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8. Tutorial

1. A Uniform Resource Locator (URL) is the address of a resource on the Internet.
 - (a) Explain the different parts of this address:
<https://www.seab.gov.sg/home/examinations/gce-a-level>
 - (i) <https://>
 - (ii) [www.](https://www.seab.gov.sg/home/examinations/gce-a-level)
 - (iii) [seab.gov.sg](https://www.seab.gov.sg/home/examinations/gce-a-level)
 - (iv) [/home/examinations/gce-a-level](https://www.seab.gov.sg/home/examinations/gce-a-level)
 - (b) Explain how the Domain Name System (DNS) resolves this URL into an IP address when entered into a web browser.
2. A voluntary organisation is considering purchasing a lease on a web domain to set up a new website to advertise their events and services. They have been advised to contact an Internet registrar. Explain the role of an Internet registrar when purchasing a domain name.
3. All Internet connected devices communicate via the TCP/IP protocol stack. This has four layers.
 - (a) Describe the roles of each layer when two devices are communicating over the Internet.
 - (b) State one hardware device commonly associated with (i) Network layer (ii) Data link layer
4. Major parts of the Internet use packet switching to transmit data between devices.
 - (a) Explain the term packet switching.
 - (b) Explain the difference between connection-oriented communication and connectionless communication.
 - (c) State one example of a protocol that uses connection-oriented communication and connectionless communication.
 - (d) A data packet contains a header and a payload. State three types of information that may be found in a packet header.
5. Explain the difference between client-server and peer-to-peer networking, and give an example of where each might be used. Include one advantage of each model.
6. Explain three advantages of using cloud-based services for organisations.



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7. A supermarket is designing a website which will allow its customers to place orders online. Customers will login to the website using email address and it will be validated during order submission.
 - (a) The individual web pages that describe each product will be generated dynamically using server-side scripting. What is a server-side scripting?
 - (b) Client and server side validation should both be done. Explain why it is important to validate the email address again once it reaches the server.
8. Explain two differences between a LAN and a WAN.
9. A company has offices in three countries connected through a WAN. Employees in each office access shared files using an internal website.
 - (a) Explain why a WAN is required rather than a LAN.
 - (b) Explain how routers help data travel between offices.
 - (c) Explain why an Intranet would be useful in this situation.
10. Explain the difference between a MAC address and an IP address.
11. Compare static and dynamic IP addresses, including how they are assigned and one advantage and disadvantage of each.
12. Explain the difference between thin client and thick client computing.
13. Explain two differences between an Intranet and the Internet.
14. Explain how a Virtual Private Network (VPN) allows secure remote access to an organisation's Intranet.
15. Explain how subnetting reduces IP address wastage in a network.
16. Explain the structure of IPv4 and IPv6 addresses, including format, size, and representation.
17. Explain the function of a router, switch, modem, and gateway in a network.